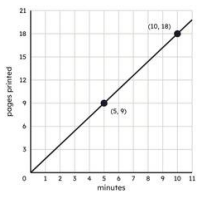
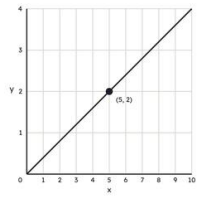
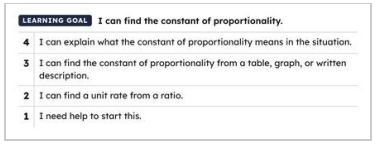
C-AR.4.2 · Weekly check · 5 questions

MA.7.AR.4.2 · Constant of proportionality · Unit 11 Representing Proportional Relationships · Given Thursday, on the week's benchmark

Form title C-AR.4.2 Check · Constant of proportionality
Description Learning goal: I can find the constant of proportionality. 5 questions, 1 point each. Work each one out on paper, then choose or type your answer. Calculator allowed.
Locked mode OFF (calculator allowed; Locked mode blocks the Chromebook calculator)

Short-answer validation (already on the template's short-answer question; copy that question): Regular expression → Matches → `^-?(\d+(\.\d+)?|\.\d+|\d+/\d+|\d+ \d+/\d+)$`
Error text: Type only a number, like 12 or -3.5 or 2/7 or 4 1/3. No units, $, %, π or commas.

#	Question image	File · type	Options to add	Answer key	Pts
1	In a proportional relationship, 64.05 cups of water are used in 7 minutes. Let x be the number of minutes and let y be the number of cups of water. What is the constant of proportionality, k, in $y = kx$? A. 64.05 B. 57.05 C. 9.15 D. 0.11	**Q1.png** Multiple choice Same problem, now multiple choice (paper: multiple choice)	A B C D	**C**	1
2	The daily fee for docking a boat at a marina is proportional to the length of the boat. The table shows the fee for four different boat lengths. Boat length (feet) | Daily fee 20 | $62.00 22 | $68.20 24 | $74.40 28 | $86.80 What is the constant of proportionality, in dollars per foot? Type a number only. No units, $ or %.	**Q2.png** Short answer	none (typed answer)	**3.1 · 3.10** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
3	The graph shows the proportional relationship between minutes and pages printed. What is the constant of proportionality, in pages per minute?  Type a number only. No units, $ or %.	**Q3.png** Short answer	none (typed answer)	**1.8 · 1.80 · 1 4/5 · 9/5** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
4	The graph shows a proportional relationship between x and y. What is the constant of proportionality? Write your answer as a fraction in simplest form.  Type a whole number, a decimal like 0.25, a fraction like 2/7 or -3/4, or a mixed number like 4 1/3 with a space after the whole number.	**Q4.png** Short answer	none (typed answer)	**2/5** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
5	Each batch of paint is made with 2 cups of blue paint and 12 cups of white paint. The relationship between cups of blue paint and cups of white paint is proportional. Let x be the number of cups of white paint and let y be the number of cups of blue paint. What is the constant of proportionality, and what does it mean in this situation? A. 2; each batch uses 2 cups of blue paint. B. 6; each cup of blue paint is used with 6 cups of white paint. C. $\frac{1}{6}$; each cup of blue paint is used with $\frac{1}{6}$ cup of white paint. D. $\frac{1}{6}$; each cup of white paint is used with $\frac{1}{6}$ cup of blue paint.	**Q5.png** Rigor Multiple choice Same problem, now multiple choice (paper: a written answer)	A B C D	**D**	1

6		**Level scale.png** Multiple choice (the template's last question)	Level 1 Level 2 Level 3 Level 4	no key	0

Answers with worked solutions are in the Answers folder next to the question images (same file names).